Лабораторная работа №2

Настройка DNS-сервера

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дисциплина: **Администрирование сетевых подсистем**

Цель работы

Целью данной работы является приобретение практических навыков по установке и конфигурированию DNS-сервера, усвоение принципов работы системы доменных имён.

root@server:~

```
  Verifying        : bind-license-32:9.16.23-34.el9_7.1.noarch
  Verifying        : bind-license-32:9.16.23-18.el9_4.1.noarch
  Verifying        : bind-utils-32:9.16.23-34.el9_7.1.x86_64
  Verifying        : bind-utils-32:9.16.23-18.el9_4.1.x86_64
  Verifying        : openssl-devel-1:3.5.1-7.el9_7.x86_64
  Verifying        : openssl-devel-1:3.0.7-27.el9.x86_64

Upgraded:
  bind-libs-32:9.16.23-34.el9_7.1.x86_64
  bind-license-32:9.16.23-34.el9_7.1.noarch
  bind-utils-32:9.16.23-34.el9_7.1.x86_64
  openssl-1:3.5.1-7.el9_7.x86_64
  openssl-devel-1:3.5.1-7.el9_7.x86_64
  openssl-libs-1:3.5.1-7.el9_7.x86_64
Installed:
  bind-32:9.16.23-34.el9_7.1.x86_64
  bind-dnssec-doc-32:9.16.23-34.el9_7.1.noarch
  bind-dnssec-utils-32:9.16.23-34.el9_7.1.x86_64
  openssl-fips-provider-1:3.5.1-7.el9_7.x86_64
  python3-bind-32:9.16.23-34.el9_7.1.noarch
  python3-ply-3.11-14.el9.0.1.noarch
```

Open ▾ *named.conf /etc Save

```
1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11         listen-on port 53 { 127.0.0.1; };
12         listen-on-v6 port 53 { ::1; };
13         directory       "/var/named";
14         dump-file       "/var/named/data/cache_dump.db";
15         statistics-file "/var/named/data/named_stats.txt";
16         memstatistics-file "/var/named/data/named_mem_stats.txt";
17         secroots-file   "/var/named/data/named.secroots";
18         recursing-file  "/var/named/data/named.recursing";
19         allow-query     { localhost; 192-168.8.8/16};
20
21         /*
22          - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23          - If you are building a RECURSIVE (caching) DNS server, you need to enable
24            recursion.
25          - If your recursive DNS server has a public IP address, you MUST enable access
```

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```
(gedit:4392): dconf-WARNING **: 12:40:35.795: failed to commit changes to dconf:
 Failed to execute child process “dbus-launch” (No such file or directory)
[root@server etc]# cp /var/named/named.loopback /var/named/master/rz/
[root@server etc]# cd /var/named/master/rz/
[root@server rz]# mv named.loopback 192.168.1
[root@server rz]# chown -R named:named /etc/named
[root@server rz]# chown -R named:named /var/named
[root@server rz]# restorecon -vR /etc
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:u
ser_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/named.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:ob
ject_r:named_conf_t:s0
[root@server rz]# restorecon -vR /var/named
[root@server rz]#
```

root@server:/var/named/master/fz

```
        type master;
        file "named.localhost";
        allow-update { none; };
};

zone "1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.ip6.arpa" IN {
        type master;
        file "named.loopback";
        allow-update { none; };
};

zone "1.0.0.127.in-addr.arpa" IN {
        type master;
        file "named.loopback";
        allow-update { none; };
};

zone "0.in-addr.arpa" IN {
        type master;
        file "named.empty";
        allow-update { none; };
};
[root@server named]# cd /var/named
[root@server named]# mkdir -p /var/named/master/fz
[root@server named]# mkdir -p /var/named/master/rz
[root@server named]# cp /var/named/named.localhost /var/named/master/fz/
[root@server named]# cd /var/named/master/fz/
[root@server fz]# mv named.localhost user.net
[root@server fz]#
```

Open

*192.168.1

/var/named/master/fz/var/named/master/rz

Save

Could not find the file "/var/named/master/fz/var/named/master/rz/192.168.1".

Please check that you typed the location correctly and try again.

```
$TTL 1D
@ IN SOA        @ server.user.net. (
        2024072700 ; seria
        1D ; refresh
        1H ; retry
        1W ; expire
        3H ) ; minimum
NS       @
        A       192.168.1.1
        PTR     server.user.net.
        $ORIGIN 1.168.192.in-addr.arpa.
        1       PTR     server.user.net.
        1       PTR     ns.user.net.
```

Open ▾ *192.168.1 Save

/var/named/master/fz/var/named/master/rz

```
$TTL 1D
@ IN SOA        @server.user.net. (
        2024072700 ; seria
        1D ; refresh
        1H ; retry
        1W ; expire
        3H ) ; minimum
        NS @
        A 192.168.1.1
        PTR server.user.net.
        $ORIGIN 1.168.192.in-addr.arpa.
        1 PTR server.user.net.
        1 PTR ns.user.net.
```

root@server:~

root@server:/var/named/master/rz | root@server:~

```
bash: lournalctl: command not found...
[root@server ~]# journalctl -x -f
Mar 04 12:55:21 server.user.net systemd[1586]: Started VTE child process 4576 la
unched by gnome-terminal-server process 2556.
    Subject: A start job for unit UNIT has finished successfully
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit UNIT has finished successfully.

    The job identifier is 679.
Mar 04 12:56:07 server.user.net sudo[4613]:  vagrant : TTY=pts/2 ; PWD=/root ; U
SER=root ; COMMAND=/bin/bash
Mar 04 12:56:07 server.user.net dbus-broker-launch[637]: avc:  op=load_policy ls
m=selinux seqno=4 res=1
Mar 04 12:56:07 server.user.net sudo[4613]: pam_unix(sudo-i:session): session op
ened for user root(uid=0) by vagrant(uid=1000)
Mar 04 12:56:07 server.user.net systemd[1]: Starting Hostname Service...
    Subject: A start job for unit systemd-hostnamed.service has begun execution
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit systemd-hostnamed.service has begun execution.
```

root@server:/var/named/master/rz

root@server:/var/named/master/rz | root@server:~

```
[root@server rz]# systemctl restart named
Job for named.service failed because the control process exited with error code.
See "systemctl status named.service" and "journalctl -xeu named.service" for det
ails.
[root@server rz]# dig ns.user.net

; <<>> DiG 9.16.23-RH <<>> ns.user.net
;; global options: +cmd
;; connection timed out; no servers could be reached

[root@server rz]# host -l user.net
;; Connection to 127.0.0.1#53(127.0.0.1) for user.net failed: connection refused
.
[root@server rz]# host -a user.net
Trying "user.net"
;; connection timed out; no servers could be reached

[root@server rz]# host -t A user.net
;; connection timed out; no servers could be reached

[root@server rz]# host -t PTR 192.168.1.1
;; connection timed out; no servers could be reached

[root@server rz]#
```

Open Save

*dns.sh
/vagrant

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install bind bind-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/dns/etc/* /etc
cp -R /vagrant/provision/server/dns/var/named/* /var/named
chown -R named:named /etc/named
chown -R named:named /var/named
restorecon -vR /etc
restorecon -vR /var/named
echo "Configure firewall"
firewall-cmd --add-service=dns
firewall-cmd --add-service=dns --permanent
echo "Tuning SELinux"
setsebool named_write_master_zones 1
setsebool -P named_write_master_zones 1
echo "Change dns server address"
nmcli connection edit "System eth0" <<EOF
remove ipv4.dns
set ipv4.ignore-auto-dns yes
set ipv4.dns 127.0.0.1
save
quit
EOF
systemctl restart NetworkManager
echo "Start named service"
systemctl enable named
systemctl start named
```

Спасибо за внимание !